



Simple Linear Models

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Optional: MLR with matrices

$$y_i = w_0 + w_1x_{i1} + w_2x_{i2} + \dots + w_px_{ip}$$

features of
sample i

parameters

$$\mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix} \begin{matrix} \text{feature 1} \\ \text{feature 2} \\ \\ \text{feature p} \end{matrix}$$

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix}$$

$$\mathbf{x}_i^T \mathbf{w} = \begin{bmatrix} x_{i1} & x_{i2} & \dots & x_{ip} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix} = w_1x_{i1} + w_2x_{i2} + \dots + w_px_{ip}$$

Optional: MLR with matrices

$$y_i = w_0 + w_1x_{i1} + w_2x_{i2} + \dots + w_px_{ip}$$

features of
sample i

parameters

$$\mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix} \begin{array}{l} \text{feature 1} \\ \text{feature 2} \\ \\ \text{feature p} \end{array}$$

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix}$$

$$\mathbf{x}_i^T \mathbf{w} = \begin{bmatrix} x_{i1} & x_{i2} & \dots & x_{ip} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix} = w_1x_{i1} + w_2x_{i2} + \dots + w_px_{ip}$$

features of
sample i

parameters

$$\mathbf{x}_i = \begin{bmatrix} 1 \\ x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix} \begin{array}{l} \\ \text{feature 1} \\ \text{feature 2} \\ \\ \text{feature p} \end{array}$$

$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix}$$

$$\mathbf{x}_i^T \mathbf{w} = \begin{bmatrix} 1 & x_{i1} & x_{i2} & \dots & x_{ip} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix} = w_0 + w_1x_{i1} + w_2x_{i2} + \dots + w_px_{ip}$$

Optional: MLR with matrices

target of sample i

features of
sample i

parameters

y_i

$$\mathbf{x}_i = \begin{bmatrix} 1 \\ x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix}$$

$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix}$$

sample 1

$$y_1 = w_0 + w_1x_{11} + w_2x_{12} + \dots + w_px_{1p} \iff y_1 = \mathbf{x}_1^T \mathbf{w}$$

$\vdots$

$\vdots$

sample i

$$y_i = w_0 + w_1x_{i1} + w_2x_{i2} + \dots + w_px_{ip} \iff y_i = \mathbf{x}_i^T \mathbf{w}$$

$\vdots$

$\vdots$

sample n

$$y_n = w_0 + w_1x_{n1} + w_2x_{n2} + \dots + w_px_{np} \iff y_n = \mathbf{x}_n^T \mathbf{w}$$

Optional: MLR with matrices

target of sample i

features of
sample i

parameters

y_i

$$\mathbf{x}_i = \begin{bmatrix} 1 \\ x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix}$$

$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix}$$

sample 1

$$y_1 = w_0 + w_1x_{11} + w_2x_{12} + \dots + w_px_{1p} \iff$$

$\vdots$

sample i

$$y_i = w_0 + w_1x_{i1} + w_2x_{i2} + \dots + w_px_{ip} \iff$$

$\vdots$

sample n

$$y_n = w_0 + w_1x_{n1} + w_2x_{n2} + \dots + w_px_{np} \iff$$

$\mathbf{y}$

$\mathbf{X}$

$$y_1 = \mathbf{x}_1^T \mathbf{w}$$

$\vdots$

$$y_i = \mathbf{x}_i^T \mathbf{w}$$

$\vdots$

$$y_n = \mathbf{x}_n^T \mathbf{w}$$

Optional: MLR with matrices

targets of
all samples

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

features of
sample i

$$\mathbf{x}_i = \begin{bmatrix} 1 \\ x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix}$$

features of all samples

$$\mathbf{X} = \begin{bmatrix} \text{---} & \mathbf{x}_1^T & \text{---} \\ \text{---} & \mathbf{x}_2^T & \text{---} \\ & \vdots & \\ \text{---} & \mathbf{x}_n^T & \text{---} \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

features of sample n

parameters

$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_p \end{bmatrix}$$

$$y_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \dots + w_p x_{ip} \quad \text{for } i = 1, 2, \dots, n$$



$$\mathbf{y} = \mathbf{X}\mathbf{w}$$

Optional: MLR with matrices

$$\min_{\hat{w}_0, \dots, \hat{w}_p} MSE(Y, \hat{Y}) = \min_{\hat{w}_0, \dots, \hat{w}_p} \frac{1}{n} \sum_{i=1}^n \left(y_i - (\hat{w}_0 + \hat{w}_1 x_{i1} + \dots + \hat{w}_p x_{ip}) \right)^2$$



$$\min_{\hat{\mathbf{w}}} \frac{1}{n} \|\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}\|_2^2$$



$$\min_{\hat{\mathbf{w}}} \|\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}\|_2^2$$

Optional: MLR with matrices

Compute the gradient of $\|\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}\|_2^2$ with respect to $\hat{\mathbf{w}}$ and set it to 0

$$\nabla_{\hat{\mathbf{w}}} \|\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}\|_2^2 = -2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}) = 0$$

$$-\mathbf{X}^T\mathbf{y} + \mathbf{X}^T\mathbf{X}\hat{\mathbf{w}} = 0$$

$$\mathbf{X}^T\mathbf{X}\hat{\mathbf{w}} = \mathbf{X}^T\mathbf{y}$$

$$\hat{\mathbf{w}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$